History of Psychology

Session 2: The birth of Psychology

Rui Mata, Center for Cognitive and Decision Sciences
October 6, 2026

Session information

Sessions take place Tuesdays, 8.15-9.45, Kollegienhaus, Aula 033. Slides will be made available shortly before each session.

#	Date	Topic	Instructor	Resources
1	22.09.2026	What kind of science is psychology?	Mata	Spektrum Podcast
2	6.10.2026	The birth of psychology	Mata	Brysbaert & Rastle (2021; 4.1-4.2)
3	13.10.2026	Psychoanalysis	Mata	Brysbaert & Rastle (2021; 4.3-4.4)
4	20.10.2026	Behaviorism	Mata	Brysbaert & Rastle (2021; 5.2)
5	03.11.2026	Gestalt and Cognitive psychology	Mata	Brysbaert & Rastle (2021; 4.2 & 5.3)
6	10.11.2026	Psychology today	Tisdall	Spear (2007)
7	17.11.2026	Psychotherapy research	Tisdall	Braakmann (2015)
8	24.11.2026	Psychological testing	Tisdall	Wasserman (2012)
9	01.12.2026	Decision science	Tisdall	Newell et al. (2022)
10	08.12.2026	What kind of science is psychology? (revisited)	Tisdall	Ball (2012)
11	15.12.2026	Exam (see below)		

Learning Objectives for Today

- Recognise that main questions in psychological science today have long traditions dating back to classical antiquity
- Identify the origins of the scientific revolution in the Renaissance and Enlightenment, and distinguish rationalist and empiricist views
- Identify key cultural/scientific movements of the 19th century and their influence on psychology, in particular 1) evolutionary theory, 2) neuropsychology, and 3) the social sciences
- Identify the 19th century as the “birth” and institutionalisation of scientific psychology
- Identify the first “schools” in Psychology - structuralism and functionalism

Roadmap for Today

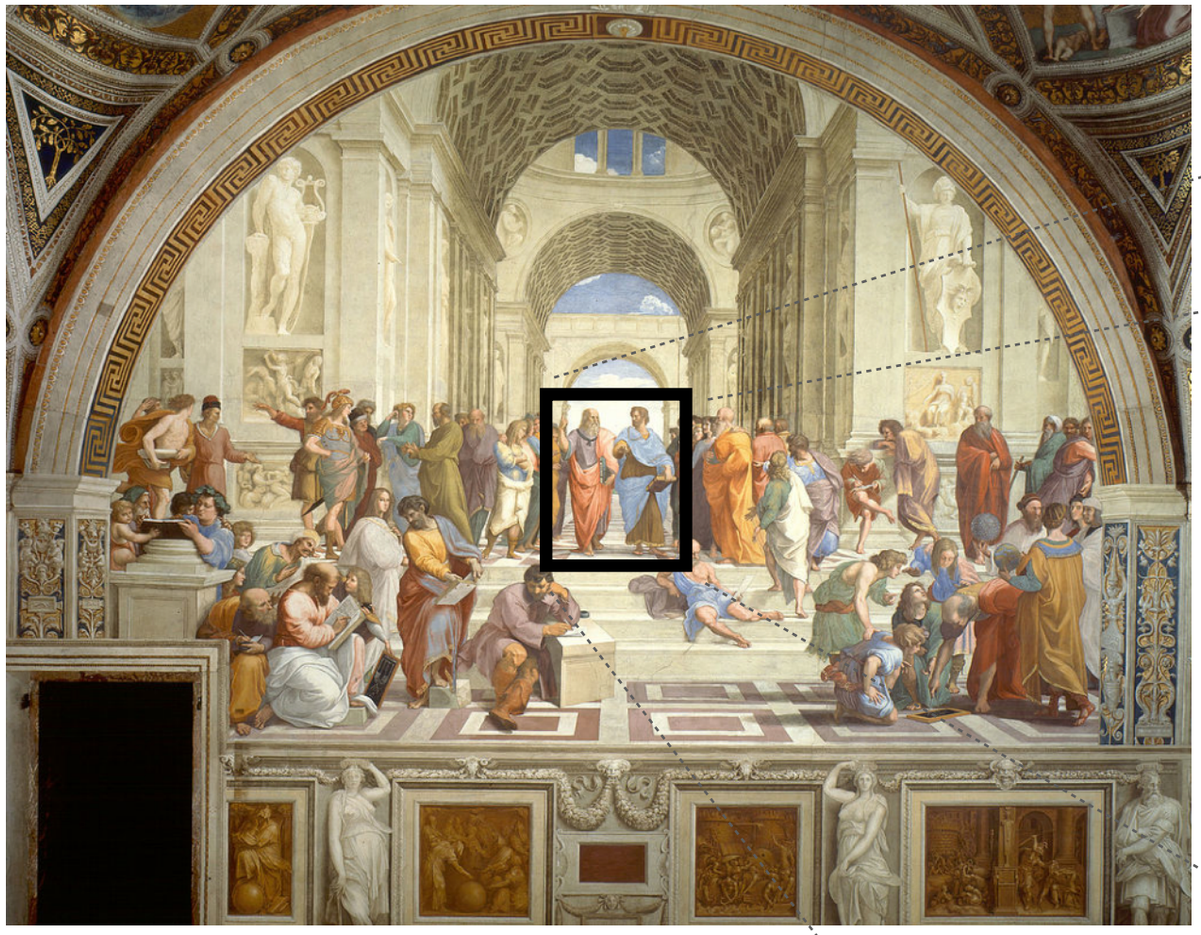
- **Ancient roots (c. 400 BC):** What is the psyche, and how can we know it?
- **Scientific Revolution & Enlightenment (1600–1800):** How should we gain knowledge: through reason or through experience?
- **19th-century currents (1830–1870):** Which ideas from biology, medicine and statistics prepared the ground?
- **The birth of scientific psychology (1879):** What made Wundt's Leipzig laboratory the "birth" of psychology?
- **The first schools (c. 1890–1910):** Structure or function: what should psychology study (first)?

- **Main idea:** *psychology's questions are ancient; its approaches and methods are varied and modern...*

Part 1 of 5: Ancient Roots



GREEK	LATIN
psukhē	psyche
breath, life, soul	*mid 17th century*



The School of Athens (1509-1511),
Raphael, Apostolic Palace, Vatican.

https://en.wikipedia.org/wiki/The_School_of_Athens



Plato
(holding Timaeus)

Aristotle
(holding Ethics)

Plato

Socrate's disciple, founded the Academy for Philosophy in 385 BC in Athens (at Academus, an olive and plane grove near Athens)

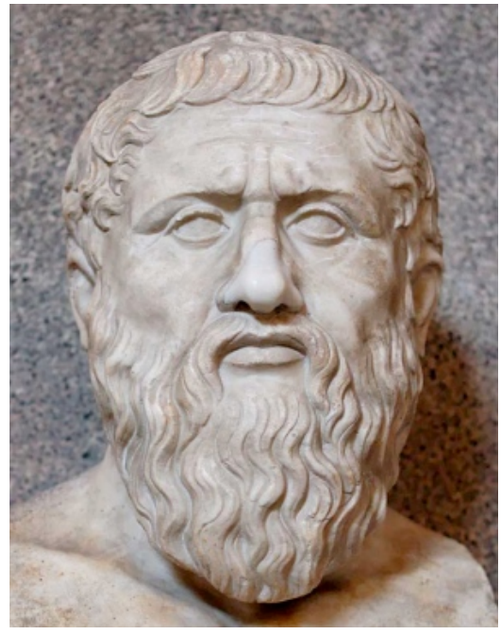
- Idealism and Theory of Forms

  the idea that reality, or reality as we can know it, is fundamentally mental or mentally constructed; senses do not necessarily deliver a true picture of the physical world (cf. analogy of the cave)

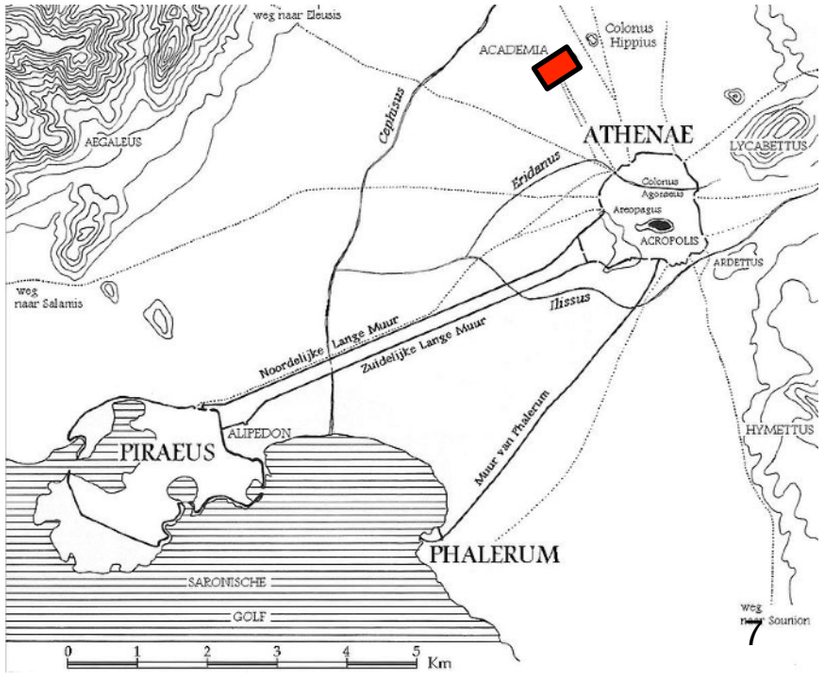
- Dualist Views (separation of body and soul)

- Nativist Views (the soul possesses a priori knowledge that can be recalled; cf. Meno's dialogue)

  https://plato.stanford.edu/entries/plato/



Plato
(427-347 BC)

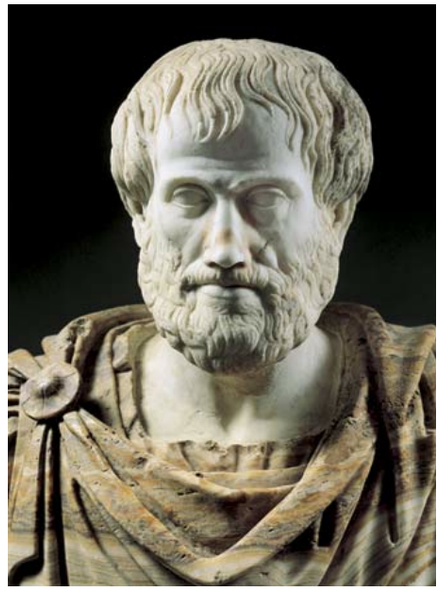


Aristotle

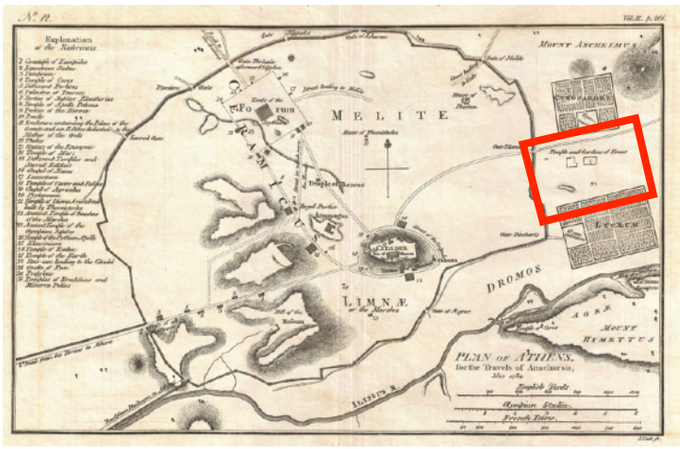
Plato's disciple, founded a philosophy school in 335 BC, the „Lyceum" at Lykeion, a grove in Athen, behind todays' Hellenic Parliament)

- Contributions to many different areas of knowledge (theoretical, practical, productive sciences)
  - biology
  - logic and causality
  - psychology: memory, dreams, health
  - emphasis on practical goals of philosophy (rhetoric, ethics, the good life)
- Monist

  body and soul are two interdependent parts that support and rely on each other; the body as a tool of the soul



Aristotele
(384-322 BC)



Bocage Map of Athens (1784)

https://plato.stanford.edu/entries/aristotle/

Take-Home: Ancient Roots

- **No "psychology" yet, but the questions are there:** the nature of knowledge and of the soul (psukhē = breath, life, soul) were central topics of Greek philosophy.
- **Plato:** idealism (reality as fundamentally mental), dualism (body and soul are separate) and nativism (knowledge is innate and recalled).
- **Aristotle:** wide empirical interests (biology, logic, memory, dreams) and monism (body and soul are interdependent).

Part 2 of 5: Scientific Revolution & Enlightenment

- Ancient roots — Plato, Aristotle
- **Scientific Revolution & Enlightenment — Galileo, Bacon**
- 19th-century currents — Darwin, Broca, Quetelet
- The birth of scientific psychology — Weber, Fechner, Wundt
- The first schools — Titchener, James

- **Guiding question:** *How should we gain knowledge: through reason or through experience?*

Scientific Revolution: Galileo

Galileo's Telescope



Galileo made his first telescope in 1609, and with it was able to, among other things, verify the phases of Venus, and discover sunspots, that he was later to use as support for a heliocentric view.

power of and need for instrumentation in science

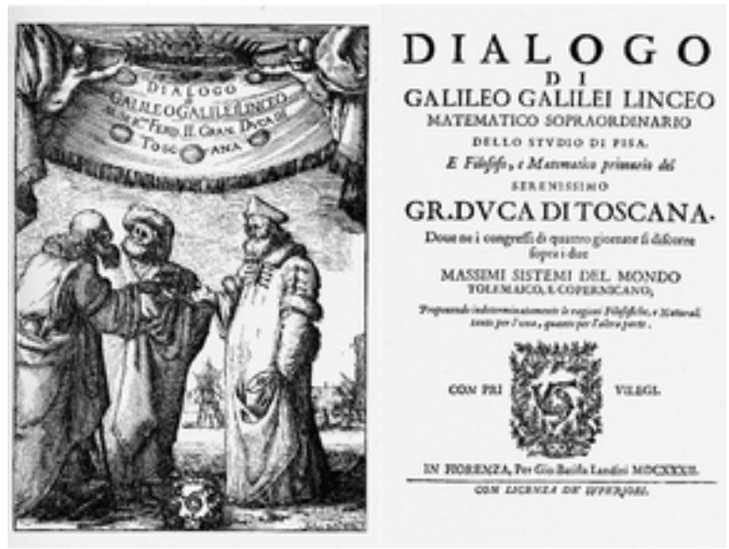
The Assayer



"Philosophy [i.e. natural philosophy] is written in this grand book — I mean the Universe — which stands continually open to our gaze, but it cannot be understood unless one first learns to comprehend the language and interpret the characters in which it is written. It is written in the language of mathematics, and its characters are triangles, circles, and other geometrical figures, without which it is humanly impossible to understand a single word of it; without these, one is wandering around in a dark labyrinth."

mathematics as the language of science

Dialogue [Concerning the Two Chief World Systems]



The book presents a series of discussions between three men, Salviati (a stand-in for Galileo), Sagredo (an initial neutral layman), and Simplicio (a follower of geocentric views). The book discusses a number of phenomena, including Venus' phases and sunspots (that are largely correct by today's standards), but also a theory of tides (that we now know to be false). The dialogue did not discuss the geocentric theory of Thyco Brahe that some astronomers preferred at the time (a hybrid system of the Copernican and Ptolemaic models; Mercury and Venus orbit the Sun, but the Sun orbits a stationary Earth; Mars, Jupiter, and Saturn orbit the Sun in much larger circles, which means they also orbit the Earth). At the time, the latter systems were not distinguishable from the existing data.

using data to distinguish theories (and science communication)

Scientific Revolution: Francis Bacon

- In 1620, his “Novum Organum Scientarium” helped establish and popularise the scientific method (i.e., inductive method). By reasoning using "induction", Bacon meant the ability to generalize a finding stepwise, based on accumulating data. He advised proceeding by this method, or in other words, by building a case from the ground up.
- For example, Bacon suggests that one draws up a list of all things in which the phenomenon to explain occurs, as well as a list of things in which it does not occur. Then one can rank the lists according to the degree in which the phenomenon occurs in each one. Then one should be able to deduce what factors match the occurrence of the phenomenon in one list and don't occur in the other list, and also what factors change in accordance with the way the data had been ranked.



Francis Bacon
(1561-1626)

https://plato.stanford.edu/entries/francis-bacon/

Scientific Revolution: Rationalism and Empiricism

Rationalism
The epistemological view that reason is the main source of knowledge, and, consequently, that there are significant ways in which concepts and knowledge can be gained independently of sensory experience — deduction.



"I think, therefore I am"

Discourse on Method (1637)

René Descartes (1596–1650)
https://plato.stanford.edu/entries/descartes/

Empiricism
The epistemological view that experience is the main source of knowledge, and that emphasizes the role of evidence in the formation of ideas and beliefs — induction—over the notion of innate knowledge or deduction.



"Let us then suppose the mind to be, as we say, white paper, void of all characters, without any ideas:—How comes it to be furnished? Whence comes it by that vast store which the busy and boundless fancy of man has painted on it with an almost endless variety? Whence has it all the materials of reason and knowledge? To this I answer, in one word, from experience."

An Essay Concerning Human Understanding (1689)

John Locke (1632-1704)
https://plato.stanford.edu/entries/locke/

Take-Home: Scientific Revolution

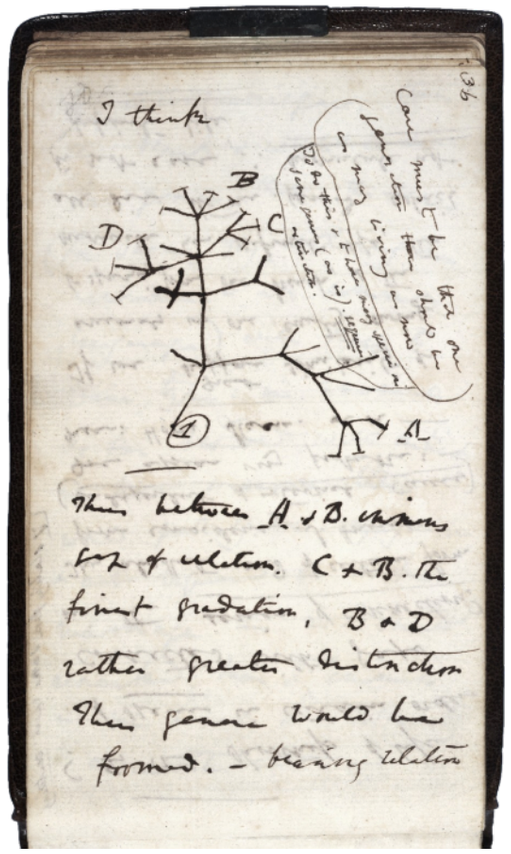
- **From authority (scholasticism) to data:** instruments, mathematics and data challenged dogma and became the way to decide between theories.
- **Induction:** The scientific approach promotes building general claims step by step from systematic observation.
- **Rationalism vs. empiricism:** philosophical discussion about whether knowledge is gained mainly through reason (deduction) or through experience (induction)? The debate still shapes psychology today.

Part 3 of 5: 19th-Century Currents

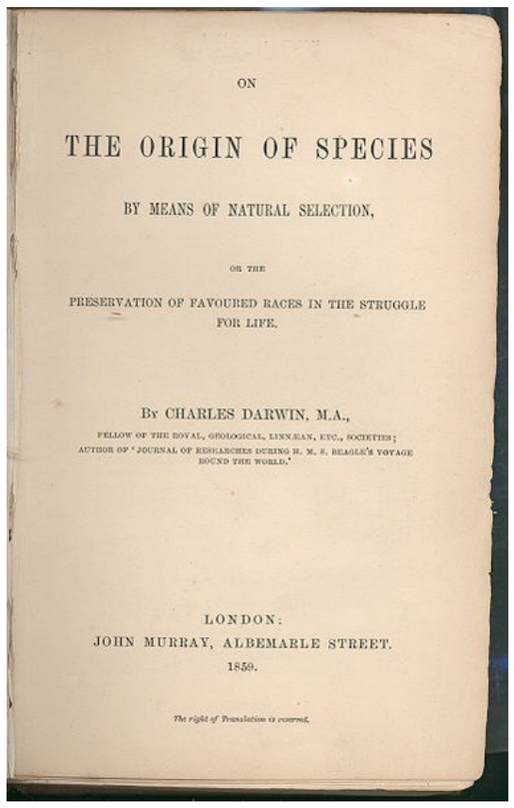
- Ancient roots — Plato, Aristotle
- Scientific Revolution & Enlightenment — Galileo, Bacon
- **19th-century currents — Darwin, Broca, Quetelet**
- The birth of scientific psychology — Weber, Fechner, Wundt
- The first schools — Titchener, James

- **Guiding question:** *Which ideas from biology, medicine and statistics prepared the ground?*

Evolutionary Theory

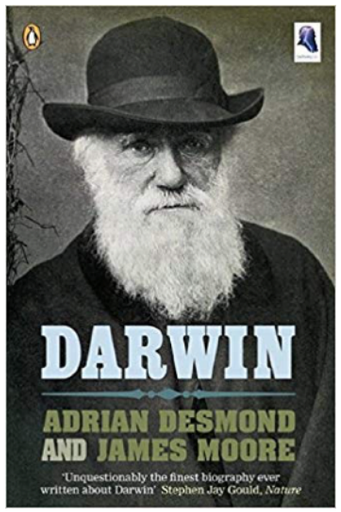


(1837)



(1859)

"In the distant future I see open fields for far more important researches. Psychology will be based on a new foundation, that of the necessary acquirement of each mental power and capacity by gradation. Light will be thrown on the origin of man and his history."



Darwin started his work on natural selection shortly after his Beagle voyage but it took him 20 years to actually publish his ideas. Desmond and Moore aim to explain this by portraying Darwin as a "closet evolutionist" that is torn between science and religion. According to Desmond and Moore, Darwin is conflicted and anxious, fearful of loosing his standing in his Anglican conservative society, and hurting his deeply religious wife. The book also makes clear how Darwin's "discovery" of natural selection is not the product of the work of an isolated genius– Darwin himself points out how **a**) Malthusian ideas about scarcity and competition (that pervaded the political discourse of the time), **b**) the concept of long periods of time having shaped the geological record (advocated by Charles Lyell), and **c**) the accumulation of facts concerning variation and similarities between species, combined to create "natural selection".

From Phrenology to Neuropsychology



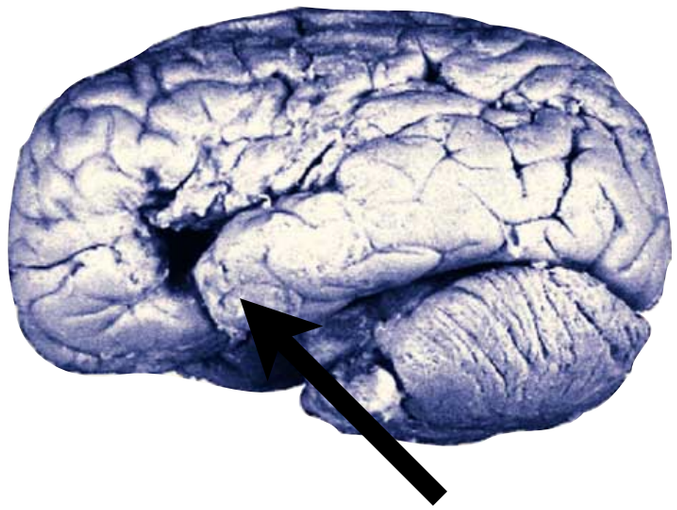
Franz Josef Gall (1758-1828). German neuroanatomist, physiologist, and pioneer in the study of the localization of mental functions in the brain. Claimed as the founder of phrenology. Gall's contributions to the field of neuropsychology were controversial at the time and now widely referred to as pseudoscience. However, Gall's study of phrenology contributed to the emergence of the naturalistic approach to the study of man, and fostered scientific inquiry into brain functions.



Pierre Paul Broca (1824-1880). French physician, anatomist and anthropologist. In 1861, Broca heard of a patient, named Louis Victor Leborgne, who had a progressive loss of speech but no other cognitive impairment. When Leborgne died just a few days later, Broca performed an autopsy. Broca went on to find autopsy evidence from additional cases in support of the localization of language articulation.

From Phrenology to Neuropsychology

The patient “Tan” (Louis Leborgne) was unable to clearly speak any words other than “tan” - and is often used as a classical example of Broca’s aphasia (expressive or non-fluent aphasia). Broca’s autopsy of Leborgne revealed an extensive lesion that involved the frontal lobe, specifically, the inferior frontal gyrus.



IFG: inferior frontal gyrus



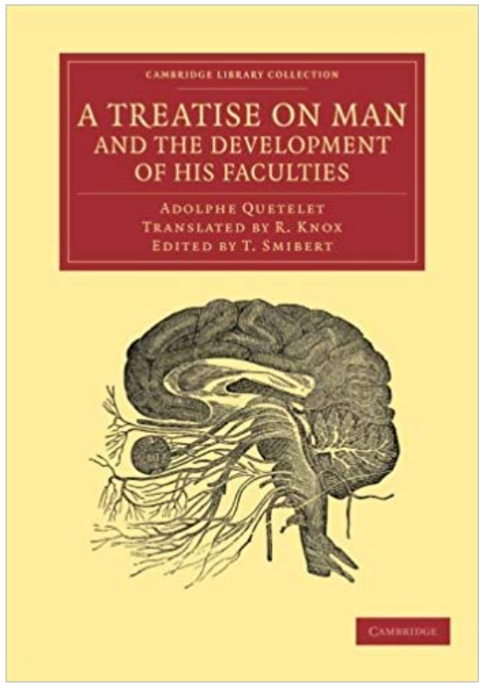
Was Broca a maverick who single-handedly changed the scientific methods of the day? Hardly. Broca’s contribution is more likely a good example of the zeitgeist that prevailed in the field at that time. The association between speech disorders and frontal lobe lesions had been already advanced by Jean-Baptiste Bouillaud in 1825. In turn, the hypothesis of left hemisphere dominance for language was first proposed by Marc Dax in 1836, also based on the observation of brain lesion patients. Broca’s methods and claims had been anticipated by over 30 years! Two conclusions can be drawn from Broca’s example. First, this episode reminds us that historical accounts often forget the contribution of separate individuals and, instead, focus on a single prominent person who can be used as a placeholder for a particular advancement or event in a field. Indeed, the naming of a brain area and aphasic type after Broca can be seen as an example of Stigler’s law of eponymy, which states that no scientific discovery is named after its original discoverer, and which emphasizes the perils of attributing any discovery to a single individual.

Cubelli, R., & De Bastiani, P. (2011). 150 Years after Leborgne: Why is Paul Broca so important in the history of neuropsychology? Cortex, 47 (2), 146–147. http://doi.org/10.1016/j.cortex.2010.11.004

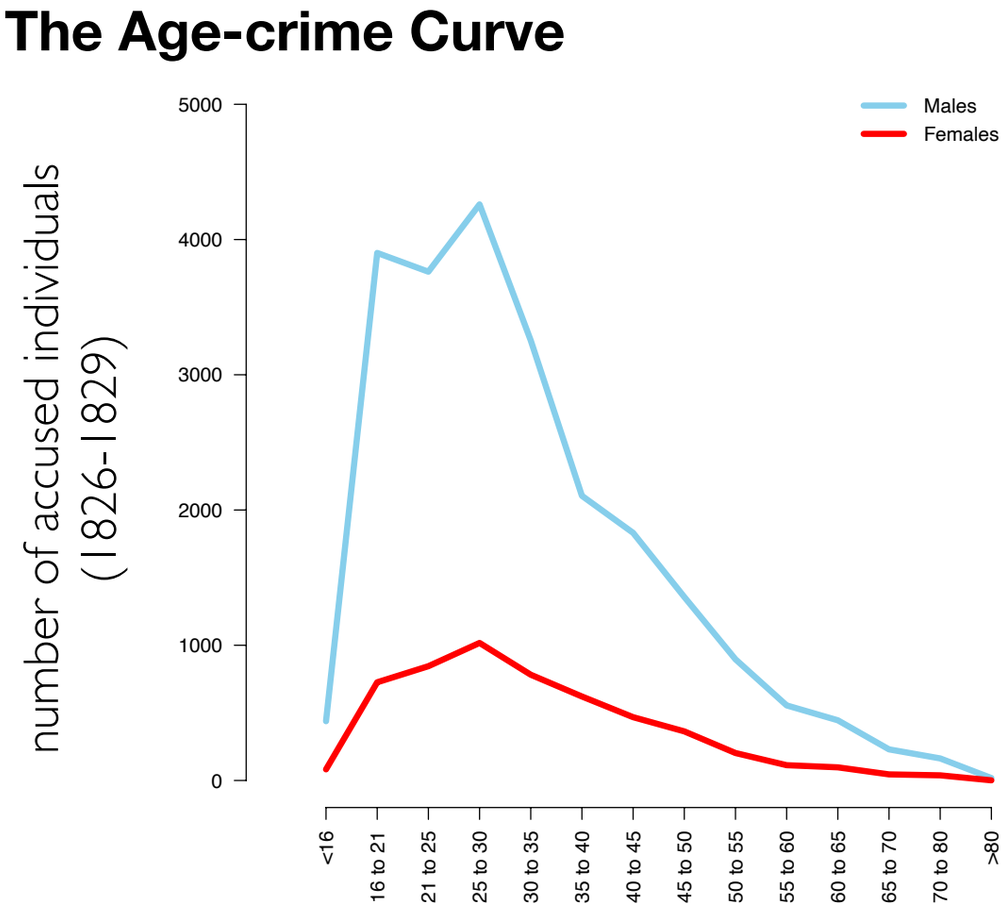
“Social physics”: Adolphe Quetelet



Adolphe Quetelet
(1796-1874)



1842



“Among all the causes which have an influence for developing or halting the propensity for crime, the most vigorous is, without contradiction, age. It is, in fact, with age that man’s **physical strength** and **passions** develop and that their energy afterwards diminishes ... This propensity must be practically nil at both extremes of life since, on the one hand, strength and passions, those two powerful instruments of crime, have scarcely been born, and when, on the other hand, their energy (pretty nearly extinguished) is found weakened by **reason** ... It is about the age of 25 years when the propensity for crime attains its maximum.”

Quetelet, A. (1831). Research on the propensity for crime at different ages.

Take-Home: 19th-Century Currents

- **Evolutionary theory:** Darwin's natural selection explains the origin of species and invites functional explanations of mind and behaviour.
- **Neuropsychology:** from Gall's phrenology to Broca's lesion studies, linking brain regions to psychological functions such as language.
- **Social physics:** Quetelet used statistics on large populations to describe regularities such as the age–crime curve.

Part 4 of 5: The Birth of Scientific Psychology

- Ancient roots — Plato, Aristotle
- Scientific Revolution & Enlightenment — Galileo, Bacon
- 19th-century currents — Darwin, Broca, Quetelet
- **The birth of scientific psychology — Weber, Fechner, Wundt**
- The first schools — Titchener, James

- **Guiding question:** *What made Wundt's Leipzig laboratory the "birth" of psychology?*

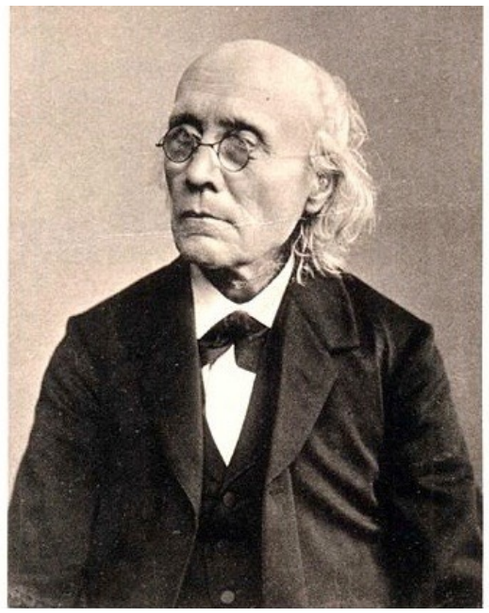
Weber and Fechner

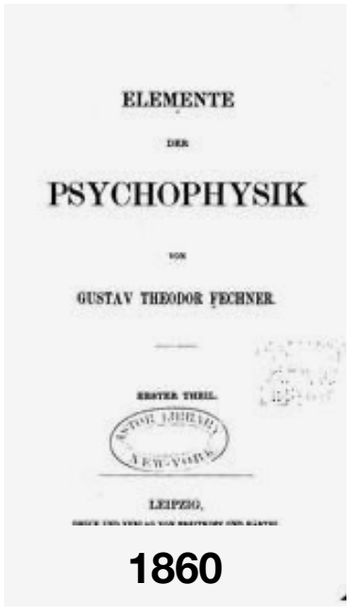


Ernst-Heinrich Weber (1795-1878)

German physician who is considered one of the founders of experimental psychology. An influential figure in the areas of physiology and psychology during his lifetime and beyond. His studies on sensation and touch, along with his emphasis on good experimental techniques gave way to new directions and areas of study for future psychologists, physiologists, and anatomists.

https://en.wikipedia.org/wiki/Ernst_Heinrich_Weber



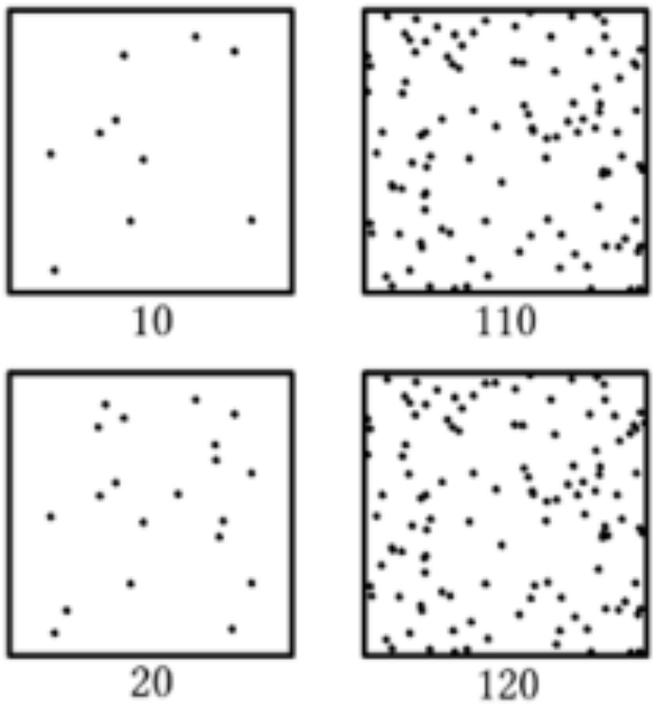


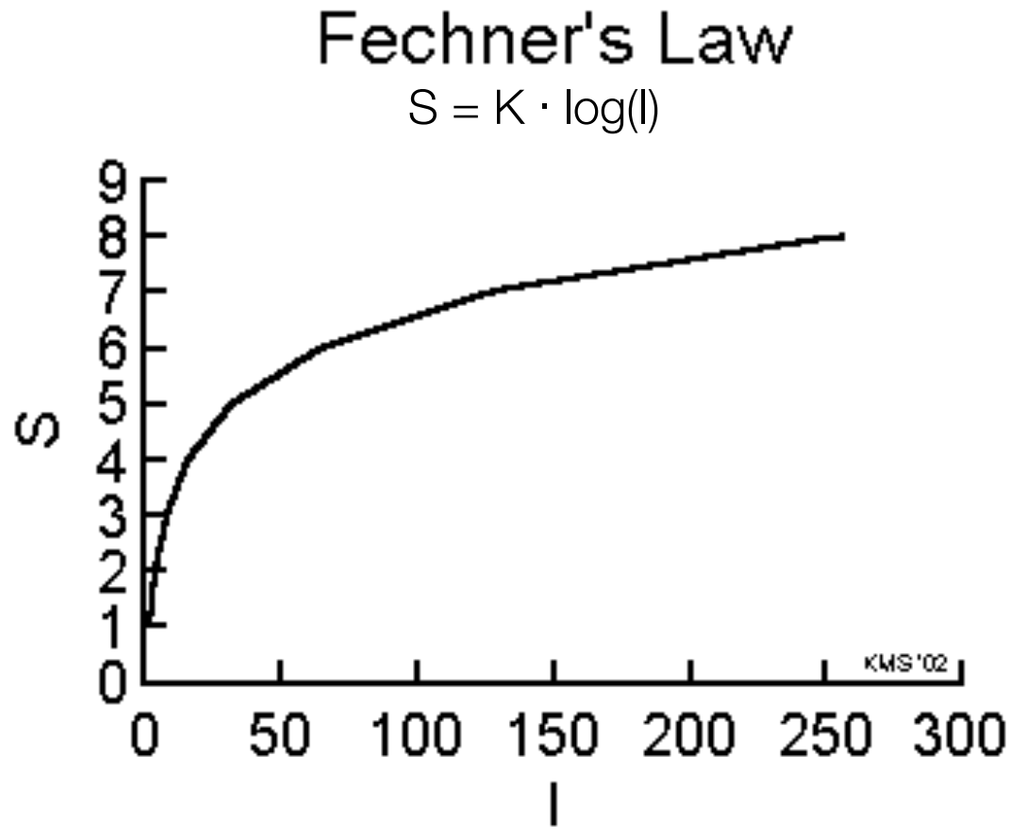
Gustav Theodor Fechner (1801-1887)

A German philosopher, physicist and experimental psychologist. An early pioneer in experimental methods applied to psychology and founder of psychophysics. He is credited with demonstrating the non-linear relationship between psychological sensation and the physical intensity of a stimulus that can be described by the formula known as the Weber–Fechner law: Sensation = K · log(Intensity).

https://en.wikipedia.org/wiki/Gustav_Fechner

Psychophysics





The figure on the left provides an illustration of the Weber–Fechner law. On each side, the lower square contains 10 more dots than the upper one. However the perception is different: On the left side, the difference between upper and lower square is clearly visible. On the right side, the two squares look almost the same. The figure on the right depicts the relation between intensity (I) of a stimulus and the subjective experience or sensation (S).

https://en.wikipedia.org/wiki/Weber–Fechner_law

Wilhelm Wundt



Wilhelm Wundt (1832-1920)

- 1832 Birth in Neckarau/Mannheim
- 1851-5 Study Medicine in Tübingen, Heidelberg, Karlsruhe
- 1856 Doctoral Degree in Heidelberg (Dr. med.)
  - Research Semester in Berlin (with Du Bois-Reymond)
- 1857 Habilitation and Privatdozentur in Heidelberg [25 years-old]
- 1858-65 Assistant to Hermann von Helmholtz in Heidelberg
- 1868(?) Prof. for Anthropology and Medical Psychology in Heidelberg
  - Member of the Badischen Fortschrittspartei (liberal)
  - Parliamentarian of the Badischen Landtags
- **1874 Professor of (Inductive) Philosophy in Zürich**
- 1875 Professor of Philosophy in Leipzig
  - **1879 Founding of the 1st. experimental Psychology Lab**
- 1917 Retirement [85 years-old!]
- 1920 Death in Großbothen, near Leipzig

https://plato.stanford.edu/entries/wilhelm-wundt/

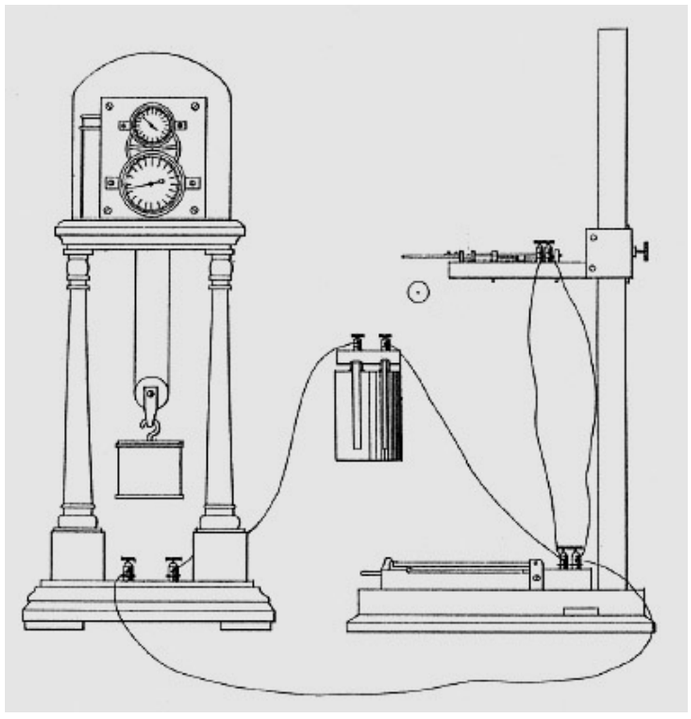
Wilhelm Wundt

„If one considers psychology as a natural science, it should be quite clear that the great revolutions that completely reshaped the physical sciences since the time of *Bacon* and *Galileo* did not have any effect upon it. For one can say with still more reason of psychology what *Kant* once remarked of logic: that it has not progressed a single step forward since *Aristotle*. (…) Those issues [e.g., nature, origin, and mortality of the soul], however, mostly do not belong to scientific psychology, but to metaphysics, because the latter, not being a natural science, has been just as unable to take any advantage of the methodological improvements of the natural sciences.”

Wilhelm Wundt, *Beiträge zur Theorie der Sinneswahrnehmung*, 1862

Instrumentation: The brass-and-glass era

Brass “instrumentation came to symbolize this stage of development, for the new psychologists were self-consciously concerned with translating perennial problems of sensation and perception into terms amenable to experimentation and measurement”



Hipp Chronoscope

A stopwatch instrument, driven by clock-work regulated by a vibrating tongue, started and stopped by a clutch activated by electro-magnets; two dials recorded duration in units down to 1,000th of a second; commonly used in reaction-time experiments. Following similar devices constructed by the English physicist Charles Wheatstone (1802 - 1875), the German clockmaker and mechanic Matthäus Hipp (1813 - 1893) developed his version in 1848. Hipp lived and worked most of his life in Bern and Neuchâtel, Switzerland. After Wilhelm Wundt (1832 - 1920) recommended the application of Hipp's chronoscope in the first edition of Grundzüge der physiologischen Psychologie in 1874, the "time viewer" was widely used in the emerging community of experimental psychologists.

Davis, R. C. (1970). The brass age of psychology. [Exhibit review]. *Technology and Culture, 11*, 604–612.

Wilhelm Wundt: Methodological eclecticism

Pluralistic Use of Methods

- **Experimental Studies**: psychophysical methods to study the connection between physical stimuli and subjective experience, reaction time measurements, accuracy of reproduction in memory tasks.
- **Introspection**: self reports of sensations, feelings, and thoughts by trained individuals under controlled experimental situations.
- **Historical and Comparative Analysis**: analysis of the products of human thought and culture through examination of different societies and their development.

Brysbaert & Rastle (2013, Chapter 4)

Take-Home: The Birth of Scientific Psychology

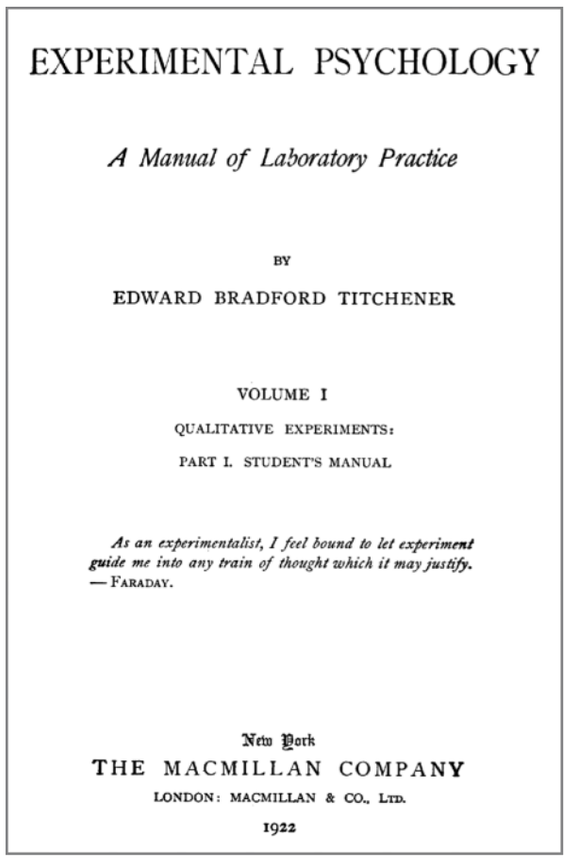
- **1879:** Wundt's Leipzig laboratory is often used as the symbolic birth of psychology as an independent, experimental science.
- **Measuring the mind:** Weber and Fechner's psychophysics ($S = K \cdot \log I$) and instrumentation (Hipp chronoscope) made mental processes quantifiable.
- **Methodological eclecticism:** Wundt combined experiments, trained introspection and historical/cultural analysis.

Part 5 of 5: The First Schools

- Ancient roots — Plato, Aristotle
- Scientific Revolution & Enlightenment — Galileo, Bacon
- 19th-century currents — Darwin, Broca, Quetelet
- The birth of scientific psychology — Weber, Fechner, Wundt
- **The first schools — Titchener, James**

- **Guiding question:** *Structure or function: what should psychology study (first)?*

Titchener and Structuralism


EXPERIMENTAL PSYCHOLOGY

A Manual of Laboratory Practice

BY

EDWARD BRADFORD TITCHENER

VOLUME I

QUALITATIVE EXPERIMENTS:

PART I. STUDENT'S MANUAL

As an experimentalist, I feel bound to let experiment guide me into any train of thought which it may justify.
— FARADAY.

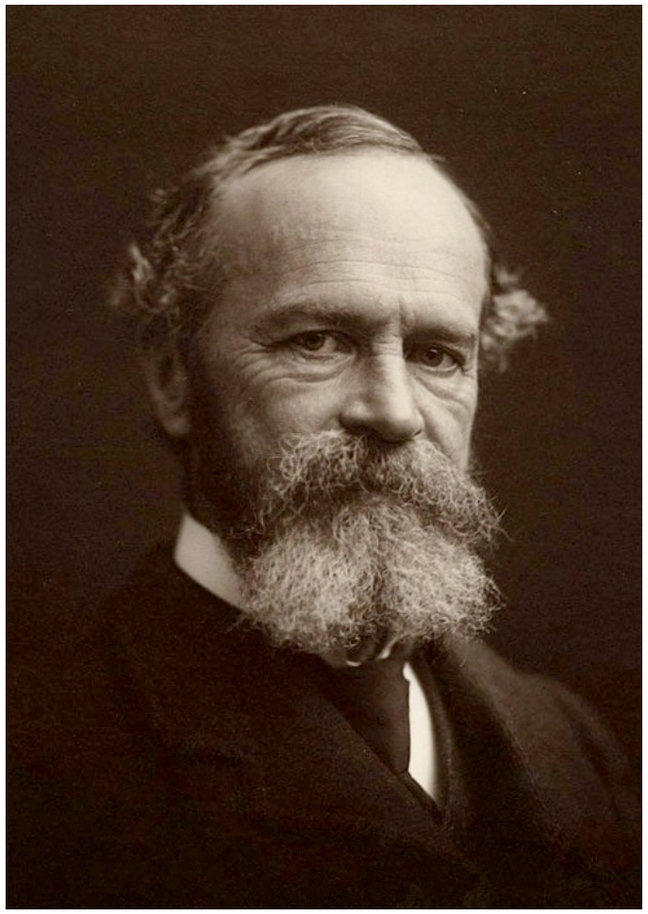
New York
THE MACMILLAN COMPANY
LONDON: MACMILLAN & CO., LTD.
1922

Titchener held that psychology is the science of consciousness and that the only method which could adequately deal with the contents of consciousness is introspection. Extensive training was required to become an investigator and subjective data had to be obtained in response to controlled stimuli. Titchener emphasized rigour in experimentation and became "the paragon of laboratory meticulousness in American psychology".

"We find a parallel to morphology in a very large portion of 'experimental' psychology. The primary aim of the experimental psychologist has been to analyze the structure of mind; to ravel out the elemental processes from the tangle of consciousness, or (if we may change the metaphor) to isolate the constituents in the given conscious formation. His task is a vivisection, but a vivisection which shall yield structural, not functional results. He tries to discover, first of all, what is there and in what quantity, not what it is there for." (…)

The postulates of a structural psychology (1898)

William James and Functionalism



William James (1842-1910)

American philosopher and psychologist who was trained as a physician. Growing up, William James attended schools in the United States, England, France, Switzerland, and Germany. He also spent time abroad in France and Germany (Dresden) before getting his doctorate. James was the first educator to offer a psychology course in the United States at Harvard in 1875. Considered to be one of the founders of functional psychology. James gained widespread recognition with his *The Principles of Psychology* (1890), totaling twelve hundred pages in two volumes, which took twelve years to complete.

"I originally studied medicine in order to be a physiologist, but I drifted into psychology and philosophy from a sort of fatality. I never had any philosophic instruction, the first lecture on psychology I ever heard being the first I ever gave".

https://plato.stanford.edu/entries/james/

The First Schools: Structuralism vs. Functionalism

"We have to consider (1) functionalism conceived as the psychology of mental operations in contrast to the psychology of mental elements; or, expressed otherwise, the psychology of the how and why of consciousness as distinguished from the psychology of the what of consciousness. We have (2) the functionalism which deals with the problem of mind conceived as primarily engaged in mediating between the environment and the needs of the organism. This is the psychology of the fundamental utilities of consciousness; (3) and lastly we have functionalism described as psychophysical psychology, that is the psychology which constantly recognizes and insists upon the essential significance of the mind-body relationship for any just and comprehensive appreciation of mental life itself."

Angell, J. R. (1907). The province of functional psychology. *Psychological Review, 14* (2). 61-91.

https://en.wikipedia.org/wiki/Structuralism_(psychology)
https://en.wikipedia.org/wiki/Functional_psychology

Take-Home: The First Schools

- **Structuralism (Titchener):** what are the elements of consciousness? Studied with rigorous, trained introspection.
- **Functionalism (James, Angell):** what is consciousness for? The mind as adaptation to the environment, inspired by evolutionary theory.
- **Shared methods:** the two schools differed more in their questions than in their tools; both relied on controlled laboratory experiments using reaction-time measurement.

Take-Home Messages for Today

- **Old questions:** psychology's central questions date back to classical antiquity.
- **New approaches:** the Scientific Revolution brought instruments, mathematics and data, and the debate between rationalism and empiricism.
- **New methods:** evolutionary theory, neuropsychology, and social statistics prepared the way for a modern psychology in the 19th century.
- **A birth date:** Wundt's Leipzig laboratory (1879) marks the institutionalisation of scientific psychology.
- **First schools:** structuralism asked what consciousness is made of; functionalism asked what it is for.